Planar universal optimality, tiered: kernel-checked consequences, a non-uniqueness theorem, and a CVODE N-body consistency check for the Yukawa interaction

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Abstract

The openai/math release contains a Lean 4 proof that the triangular lattice minimises the lower g-energy among all locally finite planar configurations of density one, for every smooth completely monotone g [1]. We file every claim we make about it in a claim ledger with the tiers of the SocrateAI-Scientific-Elenchus method [4]: **A** kernel-checked, **L** cited literature, **C** argument, **X** floating-point exploration, which may steer but may never support a claim. At Tier A, conditional only on the fidelity of upstream's definitions, we prove in Lean that the rescaled triangular lattice is universally optimal at every density in the sense of Cohn–Kumar–Miller–Radchenko–Viazovska's Definition 1.3, with upstream's lower energy, which is strictly weaker than theirs. We also prove that universal optima are *not* unique among all configurations: the lattice with one vacancy is another, and it is not an isometric image of the lattice. So the popular gloss "repelling particles always arrange into a triangular lattice" is false in this formulation. At Tier X we run an overdamped N-body relaxation for the 2D Yukawa interaction, which is in the theorem's class (Tier C), with the CVODE solver of rusty-SUNDIALS, from initial conditions derived from the Hugging Face OC20 dataset. The preregistered falsifiable prediction, that no relaxed periodic configuration falls below the triangular-lattice energy, is reported with its controls. It is consistency evidence, not proof. The ledger passes the Elenchus gate with 0 blocking findings and 6 flags, each flag marking a Tier A row whose statement adequacy no human has audited.

1 Claims and tiers

2 Tier A: what the kernel checked

Upstream (UO-A-0001). With A the triangular lattice of covolume one, $N_R(C)$ the number of points of C in the closed disk of radius R, density $\lim N_R/(\pi R^2)$, and lower energy $E_g(C) = \liminf_R N_R^{-1} \sum_{x \neq y} g(|x-y|^2)$ in $[0,\infty]$, upstream proves $\sum_{a \in A \setminus 0} g(|a|^2) \leq E_g(C)$, with equality for $C = A$, for every smooth completely monotone g. We recompiled it with the pinned toolchain (Lean 4.34.1, Mathlib `d13f23b7`, openai/math `adc7f124`, unchanged at `fd4aeeb2`); its footprint is {`propext`, `Classical.choice`, `Quot.sound`}. The proof is checked. What is not established by any gate is that these definitions are the intended ones; our item-by-item comparison with [2, Defs. 1.1–1.3] was done by AI agents and is filed as such.

Earlier corollaries (UO-A-0002–4). Riesz admissibility, finiteness of the lattice Riesz energy for $s > 2$, and optimality of $\rho^{-1/2}A$ at every density with attainment are in [3].

Ledger id	Tier	Claim (abbreviated; full statements in `ledger.json`)
UO-A-0001	A	upstream `universal_energy_minimum` (our compile and footprint; definitions AI-audited only)
UO-A-0002–4	A	Riesz admissibility and optimality; finiteness for $s > 2$; every density with attainment [3]
UO-A-0005	A	`triangular_universally_optimal`: $\rho^{-1/2}A$ is `UniversallyOptimal` for every $\rho > 0$
UO-A-0006	A	`universal_optimum_not_unique`: $A \setminus \{0\}$ is universally optimal and not an isometric image of A
UO-L-0001–4	L	CKMRV Definition 1.3; uniqueness remark after Theorem 1.4; history; composition footnote [2]
UO-C-0001	C	the 2D Yukawa interaction is completely monotone in r^2
UO-C-0002	C	the interpretation of UO-A-0006 (what it does and does not refute)
UO-X-000n	X	CVODE N-body study, controls, disk study (`claim_licence: false`)

Table 1: The claim ledger. A row may not be cited above its tier, and no row is filed above anything it rests on; both are checked mechanically, transitively, by the Elenchus gate. Every row points to a content-addressed evidence blob (compiled axiom output with source hashes, primary-source excerpts with line numbers, argument text, or result files).

Universal optimality in the sense of CKMRV (UO-A-0005). Define, in upstream's notions,

$$\begin{aligned}\mathrm{UO}(\rho, C) \;:\Longleftrightarrow\; & C \text{ locally finite}, \quad \mathrm{dens}(C) = \rho, \quad \text{and} \\ & \forall g \text{ admissible}, \ \forall C' \text{ locally finite of density } \rho : \ E_g(C) \le E_g(C').\end{aligned}$$

This transcribes [2, Def. 1.3] with one difference: CKMRV also require the energy of C to exist as a limit, and UO uses only the $\liminf$, so it is strictly weaker.

Theorem 1 (`triangular_universally_optimal`)**.** *For every $\rho > 0$,* $\mathrm{UO}(\rho, \rho^{-1/2}A)$.

Non-uniqueness (UO-A-0006).

Theorem 2 (`vacancy_energy`, `vacancy_not_isometric_image`, `universal_optimum_not_unique`)**.** *For every admissible g, $E_g(A \setminus \{0\}) = \sum_{a \in A\setminus 0} g(|a|^2)$. Hence* $\mathrm{UO}(1, A \setminus \{0\})$*, and there is no isometry f of the plane with $f(A) = A \setminus \{0\}$.*

Proof idea. Removing one point removes a sub-sum from each disk energy, and the normalisation changes by $N_R/(N_R - 1) \to 1$; the $\liminf$ comparison closes with `ENNReal.liminf_mul_le`, and upstream's theorem gives the reverse inequality. For the isometry statement, isometries of the plane are affine (Mathlib's `Isometry.affineIsometryOfStrictConvexSpace`), so $f(2a-b) = 2f(a) - f(b)$; taking preimages of $p(1,0)$ and $p(2,0)$ produces $0 \in f(A)$, contradicting $0 \notin A \setminus \{0\}$.

This is the remark after [2, Thm. 1.4] (UO-L-0002), now as a theorem in this setting. Its interpretation (UO-C-0002) is Tier C: in the lower-energy formulation over *all* locally finite configurations, "the minimiser is the triangular lattice" is false, and only the minimum value is pinned down. Uniqueness among *periodic* configurations, which CKMRV prove in dimensions 8 and 24, is open in dimension 2 and is not addressed here.

Gates. A clean sequential rebuild of the whole chain exits 0. All 22 audited declarations have footprints within the allowlist, and two planted controls (`sorry`, a custom axiom) are flagged by the same audit. Each layer was re-verified by a separate instance of the same AI model that recompiled it and wrote its own controls. That is producer≠verifier, not human review.

3 Tier L: what is known

Per CKMRV's own text (UO-L-0003): Ventevogel and Nijboer proved universal optimality of $\mathbb{Z}$ in dimension 1 (Cohn and Kumar gave another proof); before CKMRV no universal optimum was proved in any dimension > 1; CKMRV proved it for E_8 and the Leech lattice; and universal optimality of the hexagonal lattice "in full generality is an important open problem". Statements that the problem was previously solved only in dimensions 8 and 24 omit dimension 1.

4 Tier C: why the Yukawa interaction is in the class

$V(r) = e^{-\kappa r}/r$ is $g(t) = e^{-\kappa\sqrt{t}}/\sqrt{t}$ with $t = r^2$. Here $t^{-1/2}$ is completely monotone; $e^{-\kappa\sqrt{t}} = p \circ q$ with $p(u) = e^{-\kappa u}$ completely monotone and $q'(t) = \frac{1}{2}t^{-1/2}$ completely monotone, hence completely monotone by [2, footnote 2] (UO-L-0004); and products of completely monotone functions are completely monotone (Leibniz). So upstream's theorem, if its definitions are faithful, predicts that no density-one configuration has Yukawa energy below the triangular lattice's. This argument is not kernel-checked (UO-C-0001), and so the numerics below rest on Tier C, not on Tier A.

5 Tier X: a CVODE N-body consistency check

Instrument. rusty-SUNDIALS [5], crate `qf-yukawa-crystal`. $N = 288$ particles in a periodic rectangle that holds the triangular lattice 16×18 exactly at density 1; shifted-force Yukawa with cutoff $r_c = 8$ (minimum image), $\kappa = 2$ (truncation size 1.4×10^{-8}) and $\kappa = 1$ (4.2×10^{-5}). The truncated potential is not exactly completely monotone beyond r_c; that is a disclosed approximation. Overdamped relaxation $\dot{x} = -\nabla E$ is integrated by CVODE (BDF) with the analytic Jacobian $-\nabla^2 E$ until $\max|F| < 10^{-9}$, and minima are checked by the Hessian spectrum (cyclic Jacobi). Two packages named in common descriptions of this pipeline were not used: rusty-SUNDIALS has no KINSOL port, and its IDA crate documents itself as a fixed-step prototype. Annealing is overdamped Langevin (Euler–Maruyama) outside SUNDIALS, because a stochastic equation is not an ODE, followed by a CVODE quench.

Initial conditions. Twenty disordered configurations derived from the Hugging Face dataset `colabfit/OC20_IS2RES_val_id` [6] (CC-BY-4.0): random OC20 structures, rotated uniformly in 3D, projected, rescaled and accepted with a minimum separation. Twenty uniform random configurations serve as a second family. We do not claim that either is more realistic; the dataset supplies disorder, not physics. Machine-learned interatomic potentials were deliberately not used: they model atomic chemistry, not a completely monotone pair potential, and cannot test this theorem.

Preregistration. The predictions and controls were committed (`636c514`) before the solver code (`968b0f2`). P1: no relaxed configuration has energy per particle below e_{tri} by more than 10^{-9} relative. P2: at least one annealed run per family reaches the perfect lattice. P3: $T = 0$ quenches mostly end in defected local minima, which is a property of gradient flow and not evidence against the theorem.

Controls. References: $e_{\text{tri}} = 0.3935333668$ and $e_{\text{sq}} = 0.4009647699$ at $\kappa = 2$ ($e_{\text{sq}} - e_{\text{tri}} = 7.43 \times 10^{-3}$); the periodic box reproduces the direct lattice sum to 2.1×10^{-14}. A perfect lattice stays put (relative gap 5.2×10^{-14}); a lattice with $0.1a$ noise relaxes back (gap 2.5×10^{-14}); the Hessian at the lattice has exactly two zero translation modes and smallest other eigenvalue 0.0236. Negative controls: the square lattice is a stationary point with 72 negative Hessian eigenvalues (a

saddle) and relaxes below its energy; in a box that cannot hold the triangular lattice the relaxed states keep defects and stay above e_{tri} (pass).

Results. Table 2. **P1** (the falsifiable prediction): pass at $\kappa = 2$, with 0 undercuts in 100 relaxations, and pass at $\kappa = 1$, with 0 undercuts in 50. The lowest annealed gaps, 7.6×10^{-14} (OC20-derived) and 8.5×10^{-14} (uniform) at $\kappa = 2$, are the lattice itself to round-off. **P2**: pass at $\kappa = 2$, but narrowly: 1 of 20 (OC20-derived) and 1 of 20 (uniform) annealed runs reached the perfect lattice. At $\kappa = 1$ P2 *fails as preregistered*: 0 of 10 and 2 of 10. The other annealed runs end in defected or strained local minima above e_{tri}. The annealing schedule is a weak instrument, and this failure says nothing about the theorem in either direction. **P3**, as expected: 15 of 20 and 14 of 20 $T = 0$ quenches at $\kappa = 2$ keep defects. The remainder are defect-free but above e_{tri}; their structure was not analysed. Every run converged to $\max|F| < 10^{-9}$, and no final Hessian has a negative eigenvalue, so every end state is a local minimum. The negative control behaves as it must: in the incommensurate box every relaxed state is defected and at least 9.6×10^{-4} above e_{tri}. The two initial-condition families behave alike. The dataset supplied disorder, and nothing in these runs depends on its chemistry.

	family / stage	runs	perfect	defected	min rel. gap	max rel. gap	not conv.	Hess.< 0
$\kappa = 2$	incomm. box (ctl), anneal + quench	10	0	10	9.60×10^{-4}	1.73×10^{-3}	0	0
$\kappa = 2$	incomm. box (ctl), $T = 0$ quench	10	0	10	9.60×10^{-4}	5.67×10^{-3}	0	0
$\kappa = 2$	OC20-derived, anneal + quench	20	1	11	7.60×10^{-14}	2.53×10^{-3}	0	0
$\kappa = 2$	OC20-derived, $T = 0$ quench	20	0	15	1.84×10^{-4}	4.89×10^{-3}	0	0
$\kappa = 2$	uniform, anneal + quench	20	1	11	8.48×10^{-14}	2.39×10^{-3}	0	0
$\kappa = 2$	uniform, $T = 0$ quench	20	0	14	1.84×10^{-4}	6.04×10^{-3}	0	0
$\kappa = 1$	incomm. box (ctl), anneal + quench	5	0	5	3.09×10^{-4}	6.25×10^{-4}	0	0
$\kappa = 1$	incomm. box (ctl), $T = 0$ quench	5	0	5	3.10×10^{-4}	1.07×10^{-3}	0	0
$\kappa = 1$	OC20-derived, anneal + quench	10	0	5	5.77×10^{-5}	1.22×10^{-3}	0	0
$\kappa = 1$	OC20-derived, $T = 0$ quench	10	0	8	5.77×10^{-5}	1.70×10^{-3}	0	0
$\kappa = 1$	uniform, anneal + quench	10	2	3	1.93×10^{-14}	8.51×10^{-4}	0	0
$\kappa = 1$	uniform, $T = 0$ quench	10	1	8	3.96×10^{-14}	1.22×10^{-3}	0	0

Table 2: CVODE relaxations. "perfect": zero defects and $|e - e_{\mathrm{tri}}| < 10^{-8}|e_{\mathrm{tri}}|$. Relative gap $(e - e_{\mathrm{tri}})/e_{\mathrm{tri}}$. "Hess.$< 0$": runs whose final Hessian has a negative eigenvalue. The incommensurate family uses a square box and is a negative control. Tier X throughout.

Open versus closed disk. $N = 300$, $\kappa = 2$, annealed and quenched: harmonic bowl (open) versus soft wall (closed). Interior defect fraction 0.086 (open) versus 0.033 (closed); edge 0.965 versus 0.602; interior mean $|\psi_6|$ 0.913 versus 0.931. The comparison is confounded: the bowl produces a non-uniform density, the closed disk compresses particles against the wall, and the two "edge" layers (within two spacings of the outermost particle) contain different numbers of particles. It describes finite-size edge effects in these two set-ups and is not a controlled measurement of either. It is not a test of whether upstream's closed-disk counting differs from an open-disk one, which is a question about measure-zero boundary terms in a limit.

6 Limitations

1. No human has audited whether upstream's definitions, or ours, express the intended mathematics; the ledger flags every Tier A row for this.

2. UO is strictly weaker than CKMRV's universal optimality (no existence of the energy limit).

3. Uniqueness among periodic configurations in dimension 2 is open and not attempted.

4. The Yukawa step is Tier C; the numerics are Tier X and support no claim.

5. Finite N, truncated potential, heuristic annealing: P2 can fail for reasons that have nothing to do with the theorem.

Reproducibility. Lean: AutoevolveAI `formal_cert/Triangular/`; LeanMaster release `v3.48.1`. Numerics: rusty-SUNDIALS branch `physics/yukawa-triangular-2026-10-10`. Ledger and evidence: `results/discovery/ledger/` (gate: `tools/ledger.py` of [4], exit 1: 16 claims, 0 blocks, 6 flags).

AI use. The proofs, numerics, ledger and text were produced by AI agents (Claude) under the author's direction; verification runs were separate instances of the same model. The author is responsible for the content.

References

[1] OpenAI, Universal optimality of the triangular lattice, OpenAI Math Release preprint, 2026-09-23, `https://github.com/openai/math`.

[2] H. Cohn, A. Kumar, S. D. Miller, D. Radchenko, M. Viazovska, Universal optimality of the E_8 and Leech lattices and interpolation formulas, Ann. of Math. 196 (2022) 983–1082, doi:10.4007/annals.2022.196.3.3.

[3] X. Callens, Riesz-energy, finiteness and every-density corollaries of the openai/math planar universal-optimality theorem, kernel-checked in Lean 4, Zenodo (2026), doi:10.5281/zenodo.23285631.

[4] X. Callens, SocrateAI-Scientific-Elenchus, `https://github.com/xaviercallens/SocrateAI-Scientific-Elenchus` (commit `c9ae8cd`).

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[6] L. Chanussot et al., OC20 IS2RES val id, ColabFit (2024), doi:10.60732/b4005525; Open Catalyst 2020, ACS Catal. 11 (2021) 6059, doi:10.1021/acscatal.0c04525.